

PANKAJ KUMAR

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PROFESSIONAL SUMMARY

GATE 2026 qualified (AIR 6123) Computer Science undergraduate with hands-on experience building secure, production-grade full-stack applications. Engineered a civic-tech voting platform integrating AI-based biometric verification, end-to-end PII encryption, and multi-role workflow automation. Proficient in React, Node.js, TypeScript, and PostgreSQL; experienced in cross-platform mobile development with React Native.

SKILLS

Full-Stack Development: React.js, React Native, Node.js, Express.js, MongoDB, Redux, Context API

Programming Languages: JavaScript, Python

Technical Skills: Git, Github, Socket.io, Render, Vercel, Rest APIs, JWT Authentication

Soft Skills: Teamwork, Problem Solving, Effective Communication

PROJECTS

VerifiedVote — Secure Remote Voting Platform [Live Demo](#)

May 2026 – Jun 2026

- Architected a production-grade full-stack civic-tech platform (React 19 + Vite, Node.js/Express/TypeScript, PostgreSQL Neon, MinIO) supporting end-to-end postal ballot digitisation for verified exceptional-case voters.
- Built a Python FastAPI AI microservice using DeepFace (VGG-Face cosine similarity) and OpenCV with passive liveness detection (LBP non-uniformity, FFT energy ratio, chrominance variance), reducing spoofing attack surface.
- Implemented end-to-end PII encryption via pgcrypto (pgp_sym_encrypt), encrypted MinIO document keys, and role-scoped JWT secrets, meeting production-level data security requirements.
- Designed a 4-stage admin approval state machine with PostgreSQL advisory locks on cron jobs, eliminating race conditions in concurrent review workflows.
- Integrated OTP authentication (bcrypt-hashed, SMS via TextBee), Cloudflare Turnstile bot protection, Zod schema validation, and rate limiting across all API surfaces.

Tech Stack: React 19, Vite, TypeScript, Node.js, Express, PostgreSQL, pgcrypto, MinIO, Python, FastAPI, DeepFace, OpenCV, Docker, JWT

FinEduGuard — Cross-Platform Finance Education App [GitHub](#)

Jul 2025 – Aug 2025

- Developed a React Native app delivering interactive personal finance simulations (budgeting, tax, investment, fraud detection) with dynamic scenario progression, achieving cross-platform parity on Android and iOS.
- Built RESTful backend services (Node.js, Express, MongoDB) with user progress tracking, supporting concurrent session handling without data loss.

Tech Stack: React Native, Node.js, Express.js, MongoDB, Render

GameVerse — Real-Time Multiplayer PWA [GitHub](#)

Jan 2025 – Feb 2025

- Engineered a real-time multiplayer Progressive Web App (Tic-Tac-Toe, Rock-Paper-Scissors) with Socket.io bi-directional events achieving sub-100 ms move latency under concurrent load.

Tech Stack: React Native, Node.js, Expo, Socket.io, Render

Wanderers Webpage — Travel Showcase [Live Demo](#)

Jul 2024

- Designed and deployed a visually rich travel showcase using React.js, Bootstrap, and Swiper.js, featuring destination carousels and a fully responsive layout optimised for all screen sizes.

Tech Stack: React.js, Bootstrap, Swiper.js, Vercel

EDUCATION

B.Tech — Computer Science Engineering

Sep 2022 – Jun 2026

Malout Institute of Management and Information Technology

Aggregate: 80%

ACHIEVEMENTS & CERTIFICATIONS

- GATE 2026 (CS/IT) — AIR 6123 | Score: 524 (top-tier national engineering entrance exam)
- 7th Place — PSB Hackathon Series (competitive team-based development challenge)
- 1st Place — College Aptitude Test
- 3rd Place — Project Presentation Competition
- React Development Certification — Solitaire Infosys (Jul 2024)